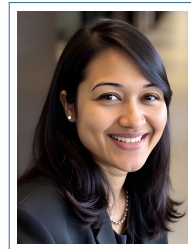


Tanzima Z. Islam

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Education

- 2013 **Ph.D., Electrical and Computer Engineering**, *Purdue University*.
Thesis: Reliable and scalable checkpointing systems for distributed computing environments.
- 2006 **B.Sc., Computer Science & Engineering**, *Bangladesh University of Engineering & Technology*.

Professional Experience

- 2026-Present **Associate Computational Scientist**, *Brookhaven National Laboratory (BNL)*, AI Department, Computing and Data Sciences Directorate.
- 2025-Present **Associate Professor**, *Texas State University (TXST)*, Department of Computer Science.
- 2020-2025 **Assistant Professor**, *Texas State University (TXST)*, Department of Computer Science.
- Summer 2024 **Visiting Faculty**, *Oak Ridge National Laboratory (ORNL)*, Computer Science and Mathematics Division (CSMD).
- Summer 2022, 2023 **Visiting Scholar**, *Brookhaven National Laboratory (BNL)*, Computational Science Initiative.
- Summer 2019 **Visiting Scholar**, *Lawrence Berkeley National Laboratory (LBNL)*, Computation Directorate.
- 2017-2019 **Assistant Professor**, *Western Washington University (WWU)*, Department of Computer Science.
- 2013-2017 **Postdoctoral Research Staff Member**, *Lawrence Livermore National Laboratory (LLNL)*, Center for Applied Scientific Computing.
Developed machine-learning techniques for performance analysis.
- 2006-2007 **Member, Research and Development**, *Commlink Info Tech Ltd.*, Dhaka, Bangladesh.
Developed software for a service-independent telecommunication network (Intelligent Network).

Awards, Honors & Inventions

- 2026 **Inaugural Genesis Mission Phase 1 Award**, Department of Energy (5% acceptance rate).
- 2025 **NSF CAREER Award**.
- 2025 **Best Paper Award**. IEEE Cloud Summit.
- 2024, 2025 **Research Millionaire Award**, College of Science and Engineering, Texas State University.
- 2023, 2025 **Presidential Distinction for Excellence in Scholarly/Creative Activities**, College of Science and Engineering (CoSE) at Texas State University.
- 2023 **Presidential Seminar Award for Excellence in Scholarly/Creative Activities** at Texas State University.
- 2023 **Valero Award for Excellence in Scholarly/Creative Activities**, Academic Affairs Council at Texas State University.
- 2022 **DOE Early Career Research Program (ECRP) Award** (10% acceptance rate).
- 2021 **College of Science and Engineering Achievement Award for Excellence in Scholarly/Creative Activities** at Texas State University.
- 2019 **R&D 100 Award**, in collaboration with Lawrence Livermore National Laboratory for inventing a scalable software solution—the Scalable Checkpoint/Restart Framework 2.0 (SCR)—that makes High Performance Computing Systems reliable and efficient.
- 2016, 2015, 2014 **Best Poster Award**, LLNL Annual Scholars Poster Symposium.

- 2014 **Lawrence Livermore National Laboratory (LLNL) Director's Science & Technology Award for Excellence in Publication.**
- 2014 2nd Place Winner, LLNL Computation Directorate Postdoctoral Poster Symposium.
- 2012, 2009 Best Paper Finalist, International Conference for High Performance Computing, Networking, Storage and Analysis (SC).
- 2010 2nd Place Winner, ACM Student Research Competition.

Research and Other Funding

Secured \$5.25M+ total in external funding as PI+Co-PI (PI only: 3.435M+).

- 2026 **PI**, OPTIX: A Multi-Modal Foundation Model for Performance-Aware HPC Code Intelligence, Department of Energy's Inaugural Genesis Mission Phase 1 Award, \$750K (9 months). Collaborators: Brookhaven National Lab, Lawrence Livermore National Lab, University of North Carolina at Chapel Hill.
- 2026 **PI**, NSF REU Supplemental Award, CRA, \$10K.
- 2025-2030 **PI**, CAREER: SPEED: Scalable Performance-aware Engine for Efficient Decision-making on HPC Systems, National Science Foundation (NSF) CAREER program, \$650K.
- Summer 2024 **PI**, Application Fingerprinting, Department of Energy's STEP: Software Tools Ecosystem Project at Oak Ridge National Laboratory, \$16K.
- Summer 2023 **PI**, SRP-HPC: Performance Analysis using Machine Learning: Use Case Chimbuko, DOE SRP Fellowship, \$85K.
- 2023-2026 **Sub-contract**, Bridging Source-Target Feature Gaps for Generalizable HPC Performance Prediction, Laboratory Directed Research and Development (LDRD) at Lawrence Livermore National Laboratory, \$310K.
- 2022-2027 **PI**, INTELYTICS: An Efficient Data-Driven Decision-Making Engine for Performance In the Era of Heterogeneity, DOE Early Career Research Program (ECRP), \$770K. (10% acceptance rate)
- 2022-2026 **Co-PI**, RECUP: Scalable Metadata And Provenance Services for Reproducible Hybrid Workflows, DOE Next Generation Data Management, \$2.43M (TXST PI: \$300K).
- 2022-2023 **Sub-contract**, ICE4HPC: Towards the Intelligent Center for HPC, Laboratory Directed Research and Development at Lawrence Livermore National Laboratory, \$100K.
- 2022 **PI**, SRP-HPC: Characterizing Workflow Applications using Machine Learning, DOE SRP Fellowship at Brookhaven National Laboratory, \$68K.
- 2021-Present **PI**, PerfROCm: Cross-platform performance prediction and analysis using deep learning, AMD Research Gift, \$225K.
- 2021-2024 **Member**, REU Site: Research Experiences for Undergraduates in Edge Computing, NSF, \$389K.
- 2021 **PI**, Predicting Performance using Few Shot Learning, Research Enhancement Program (REP) at Texas State University, \$8K.
- 2021 **Co-PI**, AMD COVID HPC Grant—equipment grant for a 5-petaflops HPC system, \$400K (estimated value).
- Summer 2019 **PI**, Proxy Application Validation for Exascale Co-design, DOE Sustainable Research Pathway Fellowship at Lawrence Berkeley National Laboratory, \$43K.
- 2018 **Co-PI**, Scientific Data Visualization course development, Office of Research and Sponsored Programs at Western Washington University, \$12K.
- 2017-2022 **PI**, Parallel Computing course, Time allocation grant from XSEDE. 100K core-hours.
- 2016 **PI**, VERITAS for Understanding Performance Evolution during Code Development, Linking Exploratory Application Research to Next-gen Development at Lawrence Livermore National Laboratory, \$200K.

Research Publications

Papers

- [1] Mohammad Zaeed, Tanzima Z. Islam, and Vladimir Indic. [Optimas: An intelligent analytics-informed generative ai framework for performance optimization](#). In *Proceedings of the 32nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining V.2*, KDD '26, page 6373–6384, New York, NY, USA, August 2026. Association for Computing Machinery. Acceptance rate: 18.5%.
- [2] Elvis G. Fefey and Tanzima Z. Islam. [Load-aware offline scheduling for heterogeneous edge deep learning inference](#). In *2026 IEEE Cloud Summit*, pages 98–103, June 2026. Acceptance rate: 34.4%.
- [3] Awatif Yasmin, Tarek Mahmud, Sana Alamgeer, **Tanzima Islam**, and Anne Ngu. [MODELX_c: A novel data-efficient personalization for fall detection](#). In *International Conference on Digital Health*. IEEE, May 2026. Accepted. (Acceptance rate: 32.4%).
- [4] **Tanzima Islam**, Holland Schutte, Mohammad Zaeed, and Aniruddha Marathe. [Data-driven analysis to understand GPU hardware resource usage of optimizations](#). *Int. J. High Perform. Comput. Appl.*, 40(3):404–420, April 2026.
- [5] Ankur Lahiry, Banooqa Banday, Yugesh Bhattarai, Mohammad Zaeed, and Tanzima Z. Islam. [Compass: A unified decision-intelligence system for navigating performance trade-off in hpc](#), April 2026.
- [6] Ashna N. Ahmed, Banooqa Banday, Terry Jones, and **Tanzima Islam**. [Attention-Informed Surrogates for Navigating Power-Performance Trade-offs in HPC](#). In *Workshop on ML for Systems in Conjunction with NeurIPS*, 2025.
- [7] Arunavo Dey, Neil Antony, Aakash Raj Dhakal, Kowshik Thopalli, Jayaraman J. Thiagarajan, Tapasya Patki, Aniruddha Marathe, Tom Scogland, Jae-Seung Yeom, and **Tanzima Islam**. [Modelx: A novel transfer learning approach across heterogeneous datasets](#). In *Proceedings of the 34th International Symposium on High-Performance Parallel and Distributed Computing*, HPDC '25, New York, NY, USA, 2025. Association for Computing Machinery. Acceptance rate: 19%.
- [8] Aakash Raj Dhakal, Tanzima Z. Islam, Arunavo Dey, Daniel Nichols, Abhinav Bhatele, Tapasya Patki, Tom Scogland, and Jae-Seung Yeom. [xAMM: "attention" to details improves cross-platform prediction accuracy](#). In *2025 IEEE 25th International Symposium on Cluster, Cloud and Internet Computing (CCGrid)*, pages 01–10, 2025. Acceptance rate: 25%.
- * [9] Elvis G. Fefey and **Tanzima Islam**. [Optimizing deep learning inference on heterogeneous devices on the edge: An ilp-based rate monotonic scheduling approach](#). In *2025 IEEE Cloud Summit*, pages 23–28, 2025. **Best Paper Award**.
- [10] Amal Gueroudji, Chase Phelps, Tanzima Z. Islam, Philip Carns, Shane Snyder, Matthieu Dorier, Robert B. Ross, and Line C. Pouchard. [Performance characterization and provenance of distributed task-based workflows on hpc platforms](#). In *Proceedings of the SC '24 Workshops of the International Conference on High Performance Computing, Network, Storage, and Analysis*, SC-W '24, page 2032–2039. IEEE Press, 2025.
- [11] Christopher Kelly, Wei Xu, Line C. Pouchard, Hubertus Van Dam, **Tanzima Islam**, Shinjae Yoo, and Kerstin Kleese Van Dam. [Performance analysis and data reduction for exascale scientific workflows](#). *The International Journal of High Performance Computing Applications (IJHPCA)*, 2025.
- [12] Matthias Maiterth, Wesley H. Brewer, Jaya S. Kuruvella, Arunavo Dey, **Tanzima Islam**, Rashadul Kabir, Kevin Menear, Dmitry Duplyakin, Tapasya Patki, Terry Jones, and Feiyi Wang. [Hpc digital twins for evaluating scheduling policies, incentive structures and their impact on power and cooling](#). In *Proceedings of the SC '25 Workshops of the International Conference for High Performance Computing, Networking, Storage and Analysis*, SC Workshops '25, page 1959–1969, New York, NY, USA, 2025. Association for Computing Machinery.
- [13] Tarek Ramadan, Nathan Pinnow, Chase Phelps, Jayaraman J. Thiagarajan, and **Tanzima Islam**. [Structure-aware representation learning for effective performance prediction](#). *Concurrency and Computation: Practice and Experience*, 37(9-11):e70046, 2025.
- [14] Raül Sirvent, Rocío Carratalá-Saez, Amal Gueroudji, **Tanzima Islam**, Line Pouchard, and Michela Taufer. [Reproducibility for hpc and distributed environments: Committees, nondeterminism, performance and workflows](#). In *ACM Conference on Reproducibility and Replicability (ACM REP)*, ACM REP '25, New York, NY, USA, 2025. Association for Computing Machinery.
- [15] Banooqa Banday, Kowshik Thopalli, **Tanzima Islam**, and Jayaraman J. Thiagarajan. [On the role of prompt construction in enhancing efficacy and efficiency of llm-based tabular data generation](#). In *2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, January 2025.

- [16] Banooqa H. Banday, Tanzima Z. Islam, and Aniruddha Marathe. [Perfgen: A synthesis and evaluation framework for performance data using generative ai](#). In *2024 IEEE 48th Annual Computers, Software, and Applications Conference (COMPSAC)*, pages 188–197, 2024. Acceptance rate: 24%.
- [17] Arunavo Dey, Aakash Dhakal, Tanzima Z. Islam, Jae-Seung Yeom, Tapasya Patki, Daniel Nichols, Alexander Movsesyan, and Abhinav Bhatele. [Relative performance prediction using few-shot learning](#). In *2024 IEEE 48th Annual Computers, Software, and Applications Conference (COMPSAC)*, pages 1764–1769, 2024. Acceptance rate: 24%.
- [18] Elvis G. Fefey and **Tanzima Islam**. [Toward efficient deep learning inference: On-node heterogeneous scheduling in edge-cloud infrastructure](#). In *2024 IEEE Cloud Summit*, pages 73–78, 2024.
- [19] Chase Phelps, Ankur Lahiry, **Tanzima Islam**, and Line Pouchard. [Reimagine application performance as a graph: Novel graph-based method for performance anomaly classification in high-performance computing](#). In *48th Annual Computers, Software, and Applications Conference (COMPSAC)*. IEEE, 2024. Acceptance rate: 24%.
- [20] Mohammad Zaeed, Tanzima Z. Islam, and Vladimir Indic. [Characterize and compare the performance of deep learning optimizers in recurrent neural network architectures](#). In *2024 IEEE 48th Annual Computers, Software, and Applications Conference (COMPSAC)*, pages 39–44, 2024. Acceptance rate: 24%.
- [21] Arunavo Dey, **Tanzima Islam**, Chase Phelps, and Christopher Kelly. [Signal processing based method for real-time anomaly detection in high-performance computing](#). In *2023 IEEE 47th Annual Computers, Software, and Applications Conference (COMPSAC)*, pages 233–240, 2023. Acceptance rate: 27%.
- [22] Bogdan Nicolae, Tanzima Z. Islam, Robert Ross, Huub Van Dam, Kevin Assogba, Polina Shpilker, Mikhail Titov, Matteo Turilli, Tianle Wang, Ozgur O. Kilic, Shantenu Jha, and Line C. Pouchard. [Building the i \(interoperability\) of fair for performance reproducibility of large-scale composable workflows in recap](#). In *2023 IEEE 19th International Conference on e-Science (e-Science)*, pages 1–7, 2023.
- [23] Chase Phelps and **Tanzima Islam**. [Automatic parallelization of cellular automata for heterogeneous platforms](#). In *2023 IEEE 47th Annual Computers, Software, and Applications Conference (COMPSAC)*, pages 352–361, 2023. Acceptance rate: 27%.
- [24] Tarek Ramadan, Ankur Lahiry, and **Tanzima Islam**. [Novel representation learning technique using graphs for performance analytics](#). In *2023 International Conference on Machine Learning and Applications (ICMLA)*, pages 1311–1318, 2023.
- [25] Line C. Pouchard, **Tanzima Islam**, and Bogdan Nicolae. [Challenges for implementing fair digital objects with high performance workflows](#). *Research Ideas and Outcomes*, 8:e94835, 2022.
- [26] Holland Schutte, Chase Phelps, Aniruddha Marathe, and **Tanzima Islam**. [LIBNVCD: An extendable and user-friendly multi-gpu performance measurement tool](#). In *2022 IEEE 46th Annual Computers, Software, and Applications Conference (COMPSAC)*, pages 73–82, 2022.
- [27] Tanzima Z. Islam and Chase Phelps. [Hpc@scale: A hands-on approach for training next-gen hpc software architects](#). In *2021 IEEE 28th International Conference on High Performance Computing, Data and Analytics Workshop (HiPCW)*, pages 29–34, 2021.
- [28] Tanzima Z. Islam, Philip Wu Liang, Forest Sweeney, Cody Pragner, Jayaraman J. Thiagarajan, Moushumi Sharmin, and Shameem Ahmed. [College life is hard! - shedding light on stress prediction for autistic college students using data-driven analysis](#). In *2021 IEEE 45th Annual Computers, Software, and Applications Conference (COMPSAC)*, pages 428–437, 2021.
- [29] Quentin Jensen, Filip Jagodzinski, and **Tanzima Islam**. [FILCIO: Application agnostic i/o aggregation to scale scientific workflows](#). In *2021 IEEE 45th Annual Computers, Software, and Applications Conference (COMPSAC)*, pages 1587–1592, 2021.
- [30] Tarek Ramadan, Tanzima Z. Islam, Chase Phelps, Nathan Pinnow, and Jayaraman J. Thiagarajan. [Comparative code structure analysis using deep learning for performance prediction](#). In *2021 IEEE International Symposium on Performance Analysis of Systems and Software (ISPASS)*, pages 151–161, 2021. Acceptance rate: 36%.
- [31] Jack Stratton, Michael Albert, Quentin Jensen, Max Ismailov, Filip Jagodzinski, and **Tanzima Islam**. [Towards aggregation based i/o optimization for scaling bioinformatics applications](#). In *2020 IEEE 44th Annual Computers, Software, and Applications Conference (COMPSAC)*, pages 1250–1255, 2020.
- [32] **Tanzima Islam**, Alexis Ayala, Quentin Jensen, and Khaled Ibrahim. [Toward a programmable analysis and visualization framework for interactive performance analytics](#). In *2019 IEEE/ACM International Workshop on Programming and Performance Visualization Tools (ProTools)*, pages 70–77, 2019.

- [33] Tapasya Patki, Jayaraman J. Thiagarajan, Alexis Ayala, and Tanzima Z. Islam. [Performance optimality or reproducibility: that is the question](#). In *Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis*, SC '19, New York, NY, USA, 2019. Association for Computing Machinery. Acceptance rate: 20%.
 - [34] Nicholas Majeske, Filip Jagodzinski, Brian Hutchinson, and **Tanzima Islam**. [Low rank smoothed sampling methods for identifying impactful pairwise mutations](#). In *Proceedings of the 2018 ACM International Conference on Bioinformatics, Computational Biology, and Health Informatics*, BCB '18, page 681–686, New York, NY, USA, 2018. Association for Computing Machinery.
 - [35] Jayaraman J. Thiagarajan, Rushil Anirudh, Bhavya Kailkhura, Nikhil Jain, **Tanzima Islam**, Abhinav Bhatele, Jae-Seung Yeom, and Todd Gamblin. [PADDLE: Performance Analysis using a Data-driven Learning Environment](#). In *IEEE International Parallel and Distributed Processing Symposium (IPDPS)*, May 21–25 2018. Acceptance rate: 24.5%.
 - [36] Teng Wang, Adam Moody, Yue Zhu, Kathryn Mohror, Kento Sato, **Tanzima Islam**, and Weikuan Yu. [Metakv: A key-value store for metadata management of distributed burst buffers](#). In *2017 IEEE International Parallel and Distributed Processing Symposium (IPDPS)*, pages 1174–1183, 2017.
 - [37] Tania Banerjee, Jason Hackl, Mrugesh Shringarpure, **Tanzima Islam**, Sivaramakrishnan Balachandar, T. Jackson, and Sanjay Ranka. [Cmt-bone — a proxy application for compressible multiphase turbulent flows](#). In *IEEE International Conference on High Performance Computing, Data, and Analytics (HiPC)*, pages 173–182. IEEE, 12 2016.
 - [38] **Tanzima Islam**, Kathryn Mohror, and Martin Schulz. [Exploring the MPI Tool Information Interface: Features and Capabilities](#). *The International Journal of High Performance Computing Applications (IJHPCA)*, 30(2):212–222, 2016.
 - [39] **Tanzima Islam**, Jayaraman J. Thiagarajan, Abhinav Bhatele, Martin Schulz, and Todd Gamblin. [A Machine-Learning Framework for Performance Coverage Analysis of Proxy Applications](#). In *International Conference for High Performance Computing, Networking, Storage and Analysis (SC)*, Salt Lake City, UT, November 13–18 2016. Acceptance rate: 23%.
 - [40] Lee Savoie, David K. Lowenthal, Bronis R. de Supinski, **Tanzima Islam**, Kathryn Mohror, Barry Rountree, and Martin Schulz. [I/O Aware Power Shifting](#). In *IEEE International Parallel and Distributed Processing Symposium (IPDPS)*, pages 740–749, May 2016. Acceptance rate: 23%.
 - [41] **Tanzima Islam**, Kathryn Mohror, and Martin Schulz. [Exploring the capabilities of the new mpi₂ interface](#). In *Proceedings of the 21st European MPI Users' Group Meeting*, EuroMPI/ASIA '14, page 91–96, New York, NY, USA, 2014. Association for Computing Machinery.
 - [42] Anup Mohan, Thomas Hacker, Gregory P. Rodgers, and **Tanzima Islam**. [Batchsubmit: A high-volume Batch Submission System for Earthquake Engineering Simulation](#). *Concurrency and Computation: Practice and Experience*, 26(13):2240–2252, 2014.
 - [43] John R Tramm, Andrew R Siegel, **Tanzima Islam**, and Martin Schulz. [XSbench - the development and verification of a performance abstraction for Monte Carlo reactor analysis](#). In *PHYSOR 2014 - The Role of Reactor Physics toward a Sustainable Future*, Kyoto, 2014.
 - [44] **Tanzima Islam**, Saurabh Bagchi, and Rudolf Eigenmann. [Reliable and Efficient Distributed Checkpointing System for Grid Environments](#). *Journal of Grid Computing (JoGC)*, 12(4):593–613, Dec 2014.
 - [45] **Tanzima Islam**, Kathryn Mohror, Saurabh Bagchi, Adam Moody, Bronis R De Supinski, and Rudolf Eigenmann. [McrEngine: A Scalable Checkpointing System Using Data-Aware Aggregation and Compression](#). *Scientific Programming*, 21(3-4):149–163, 2013.
 - * [46] **Tanzima Islam**, Kathryn Mohror, Saurabh Bagchi, Adam Moody, Bronis R. de Supinski, and Rudolf Eigenmann. [McrEngine: A Scalable Checkpointing System Using Data-aware Aggregation and Compression](#). In *International Conference on High Performance Computing, Networking, Storage and Analysis (SC)*, pages 17:1–17:11, 2012. Acceptance rate: 20%. **Best student paper finalist**.
 - * [47] **Tanzima Islam**, Saurabh Bagchi, and Rudolf Eigenmann. [FALCON: A System for Reliable Checkpoint Recovery in Shared Grid Environments](#). In *Proceedings of the Conference on High Performance Computing Networking, Storage and Analysis (SC)*, pages 1–12, Nov 2009. **Best student paper finalist**.
- Policy Reports, Book Chapters, Articles**
- [48] Lois Curfman McInnes, Dorian Arnold, Prasanna Balaprakash, Mike Bernhardt, Beth Cerny, Anshu Dubey, Roscoe Giles, Denise Ward Hood, Mary Ann Leung, Vanessa Lopez-Marrero, Paul Messina, Olivia B. Newton, Chris Oehmen, Stefan M. Wild, Jim Willenbring, Lou Woodley, Tony Baylis, David E. Bernholdt, Chris Camano, Johannah Cohoon, Charles Ferenbaugh, Stephen M. Fiore, Sandra Gesing, Diego Gomez-Zara, James Howison, **Tanzima Islam**, David Kepczynski, Charles Lively, Harshitha Menon, Bronson

Messer, Marieme Ngom, Umesh Paliath, Michael E. Papka, Irene Qualters, Elaine M. Raybourn, Katherine Riley, Paulina Rodriguez, Damian Rouson, Michelle Schwalbe, Sudip K. Seal, Ozge Surer, Valerie Taylor, and Lingfei Wu. [Report of the 2025 workshop on next-generation ecosystems for scientific computing: Harnessing community, software, and ai for cross-disciplinary team science](#), 2025.

- [49] SIAM Task Force on the Future of Computational Science. [SIAM Task Force Anticipates Future Directions of Computational Science](#). *Society for Industrial and Applied Mathematics*, June 2024.
- [50] Chris Daw, Brian Barragan Cruz, Nicholas Majeske, Filip Jagodzinski, **Tanzima Islam**, and Brian Hutchinson. Low rank approximation methods for identifying impactful pairwise protein mutations. In Nurit Haspel, Filip Jagodzinski, and Kevin Molloy, editors, *Algorithms and Methods in Structural Bioinformatics*, pages 63–87. Springer Nature, 2022.
- [51] Ritu Arora, Xiaosong Li, Bonnie Hurwitz, Daniel Fay, Dhabaleswar K. Panda, Edward Valeev, Shaowen Wang, Shirley Moore, Sunita Chandrasekaran, Ting Cao, Holly Bik, Matthew Curry, and **Tanzima Islam**. [Future directions of the cyberinfrastructure for sustained scientific innovation \(cssi\) program](#), 2020.
Ph.D. Dissertation
- [52] **Tanzima Islam**. [Reliable and scalable checkpointing systems for distributed computing environments](#). PhD thesis, Purdue University, West Lafayette, IN, May 2013.

Selected Open-Source Software & Dataset Releases

GPTune	Integrated the interpretable machine learning algorithm for identifying important performance-influencing factors—developed as part of the Dashing framework—into GPTune, an auto-tuner developed at Lawrence Berkeley National Laboratory.
SCR	Integrated a scalable data compression method—mcrEngine—into Scalable Checkpoint/Restart library (SCR) for MPI applications in collaboration with LLNL. SCR will serve as the reliability backbone for LLNL's upcoming exascale system, El Capitan.
libNVCD	Easy-to-use, performance measurement and analysis tool for NVIDIA based GPUs.
DASHING	Interpretable machine learning algorithms for performance characterization and comparison.
Gyan	Performance measurement tool for MPI libraries in collaboration with LLNL.
PyPerfDump	Hardware performance counter measurement tool for Python applications.
Dataset 1	On-node scaling data on HPC systems. doi:10.5281/zenodo.4315003 .
Dataset 2	Performance characterization data for several DOE Exascale Computing Project (ECP) applications including Particle-In-Cell application WarpX and cosmological simulation code Nyx3D. doi:10.5281/zenodo.3403037 .

Selected Professional Activities

Organizer	Chair → General Co-chair@IPDPS'27, Student Program@SC'27, Posters@ISC'26, Performance Track@SC'26, Software Track@ICPP'26, Performance Track@IPDPS'25, Applications and Workflows Track@CCGrid'25, Posters@HPDC'25, Performance Track@SC'22; Organizer → IEEE International Workshop on Big Data Computation, Analysis & Applications (in conjunction with IEEE COMPSAC'19–'23); Vice/Co-Chair → Posters@ISC'25, Performance Track@HiPC'24, Awards@SC'23, Performance Track@IEEE Cluster'23, Performance Track@ICPP'19.
Selected Technical Program Committees	NeurIPS'22, '26; SIGKDD'26; High-Performance Distributed Computing (HPDC)'22–Present; International Parallel & Distributed Processing Symposium (IPDPS)'17–Present; International Conference for High Performance Computing, Networking, Storage and Analysis (SC)'17–Present (Performance track); SC'19–'21 (Posters); SC'18 (HPC for Undergraduates); SC'15 (Resilience BoF); International Supercomputing Conference (ISC)'19–Present; Women in HPC Technical Conference'20; EuroMPI User Forum'20; Grace Hopper Celebration (GHC)'19 (Computer Systems Engineering track); The Platform for Advanced Scientific Computing (PASC)'19; International Conference on Parallel Processing (ICPP)'18–Present; International Conference on Distributed Computing Systems (ICDCS)'17 (Workshop); International Symposium on Computer Architecture and High Performance Computing (SBAC-PAD)'13–'14; Symposium on Principles and Practice of Parallel Programming (PPoPP)'11; Dependable Systems and Networks (DSN)'10, '22.
Editor	Subject matter editor for Future Generation Computer Systems (FGCS)'25; Parallel Computing (ParCo)'25–Present.
Advisor/Mentor	CAHSI NAIRR Grant Program.
Technical Board	Technical review board member at IEEE Transactions on Parallel and Distributed Systems (TPDS)'20–Present.

Grant Review Committees	DOE ECRP; DOE SciDAC; DOE SBIR/STTR; NSF FIRE; NSF ACSS (Category II); NSF Cybertraining/OAC; NSF CISE/CSSI; NSF CISE/SHF; NSF CISE/OAC; NSF CISE/CCF; NSF HBCU-UP.
Selected Journal Reviewing	IEEE Transactions on Parallel and Distributed Systems (TPDS)'19–Present; Parallel Computing (ParCo)'19–'21; Journal of Grid Computing (JoGC)'17–'21; International Journal of High Performance Computing Applications (IJHPCA)'18; Concurrency and Computation: Practice and Experience (CCPE)'16–'17.
Memberships	IEEE, ACM.

Invited Talks and Panels

September, 2026	What Decisions Should HPC Make So You Do Not Have To? , <i>CCDSC</i> , Lyon, France, Invited by Turing Award winner Dr. Jack Dongarra.
June, 2026	From Modeling to Decision Intelligence: Learning to Navigate Power–Performance Trade-offs in HPC Systems , <i>Women in HPC (WHPC) at the International Supercomputing Conference (ISC)</i> , Hamburg, Germany.
June and August, 2025	From Reactive Debugging to Proactive Detection: AI/ML for Performance-Aware Software Development , <i>Monterey Data Conference (invite-only DOE workshop)</i> , Monterey, CA; <i>PASC</i> , Switzerland; <i>EuroPar</i> , Germany.
September, 2024	Reimagine Performance Modeling: Challenges and Opportunities with Generative AI , <i>CCDSC</i> , Lyon, France, Invited by Turing Award winner Dr. Jack Dongarra.
November, 2024	Women in HPC Panel , <i>San Diego Supercomputer Center panel at the International Conference for High Performance Computing, Networking, Storage and Analysis (SC)</i> , Atlanta, GA.
2024	Reimagine Performance Modeling: Challenges and Opportunities with Generative AI (expanded) , <i>GCLab, University of Tennessee</i> , Knoxville, TN.
August, 2024	Augmenting AMD's Omni-perf with ML-based In-depth Influence Analysis for Explaining Performance Gap , <i>Coral-2 Meeting at Lawrence Livermore National Laboratory (LLNL)</i> , Virtual.
2024, 2025	Intelytics: An Efficient Data-Driven Decision-Making Engine for Performance In the Era of Heterogeneity , <i>DOE ASCR PI Meeting</i> , Atlanta, GA and Texas, respectively.
2023	Building Your Research Brand , <i>Rising Stars Panel at the University of Texas at Austin</i> , Austin, TX.
September, 2022	How to Apply to Graduate School: Things I Wish I Knew , <i>Bangladesh University of Engineering and Technology (BUET)</i> , Virtual.
May, 2022	How to be a Great Mentor , <i>Exascale Computing Project (ECP)</i> , Virtual.
April, 2022	Scalability Challenges and Opportunities for I/O Bound Applications , <i>CHEOPS Workshop at EuroSys</i> , Virtual, https://tinyurl.com/2xyb2wnr .
March, 2022	Exascale Computing Project (ECP): Enabling Next-Generation of Hardware-Software Co-design using Data Science , <i>Lawrence Livermore National Laboratory (LLNL)</i> , Virtual.
February, 2022	Invited Seminar , <i>Sustainable Horizons Institute (SHI)</i> , Virtual.
December, 2021	Characterizing Performance of Workflow Applications , <i>Brookhaven National Laboratory (BNL)</i> , Virtual.
December, 2021	Center of Excellence: Performance Characterization of Deep Learning Workloads , <i>AMD</i> , Austin, TX.
September, 2021	Enabling Next-Generation Software and Co-Design , <i>AMD at Texas State University</i> , San Marcos, TX.
September, 2021	HPC Enables Digitalization Panel , <i>Digital 360 Summit</i> , San Marcos, TX.
November, 2020	Careers in HPC Panel—Taking the Leap: Changing Careers , <i>International Conference for High Performance Computing, Networking, Storage and Analysis (SC)</i> , Virtual.
October, 2020	May the HPC Force be with You! , <i>Richard Tapia Celebration of Diversity in Computing</i> , Virtual, Organized a panel introducing undergraduate students from underserved and underrepresented communities to HPC.
August, 2020	Data-Driven Performance Modeling for HPC Co-design , <i>AMD</i> , Virtual, Invited scholar's talk.
August, 2020	Dashing: An Interpretable Machine Learning Toolkit for Performance Analysis and Visualization , <i>Exascale Computing Project Hackathon at Oak Ridge National Laboratory and Brookhaven National Laboratory</i> , Virtual.
May, 2019	Scalable I/O Performance for Scientific Applications—Challenges and Potentials , <i>NIMBioS Workshop on Scientific Collaboration Enabled by High Performance Computing</i> , Knoxville, TN.
December, 2018	Understanding the Performance Portability Challenges and Opportunities using Machine Learning , <i>Lawrence Berkeley National Laboratory (LBNL)</i> , Berkeley, CA.
November, 2018	HPC for Undergraduates , <i>International Conference for High Performance Computing, Networking, Storage and Analysis (SC)</i> , Dallas, TX.

- July, 2018 **Machine Learning Framework for Performance Coverage Analysis of Proxy Applications**, *The Platform for Advanced Scientific Computing Conference (PASC)*, ACM and the Swiss National Supercomputing Centre (CSCS), Basel, Switzerland.
- November, 2015 **How HPC Shaped My Career**, *International Conference for High Performance Computing, Networking, Storage and Analysis (SC)*, Austin, TX.
- February, 2015 **Proxy Application Validation using Machine Learning (Veritas)**, *JOWOG-34*, Sandia National Laboratories, Albuquerque, NM.
- January, 2014 **Opportunities after Graduate School**, *Spelman College*, Atlanta, GA.

Teaching Experience

- Sprg'21-'24 HPC@Scale
- Summer'22 Data Science Pathways (online certification program)
- Fall'21, Fall'23 Linear Data Structures (CS-II)
- Sprg'20, Fall'20-'22 Compiler Construction
- Fall'18, Sprg'19 Graduate Parallel Computing
- Win'18, Sprg'19 Graduate/Undergraduate Parallel Computing
- Fall'18 Scientific Data Visualization (co-designed and co-taught)
- Sprg'18, Win'19 Computer Networks
- Fall'17-'18, Sprg'18, Win'19 Advanced Data Structures

Outreach and Mentoring Students

- 2014-Present **Co-founder**, *Bangladeshi Women in Computer Science and Engineering*, Dhaka, Bangladesh, First research, networking, and mentoring platform for female Computer Science and Engineering students of Bangladesh. Mentored 100+ female students through graduate applications, fellowship and scholarship submissions, research and internship preparation, career planning, and leadership development in computing.
- 2017-Present **Research advisor**, *Undergraduate (10), M.Sc. (5), Ph.D. (8), and Postdoc (1) (TXST); Undergraduate (7) and graduate (10) students (WWU)*, High school (2); K-5 AI (4).
- 2019-Present **Mentor**, *REU (3), Graduate students (10)*, SC Student Volunteer Program.
- 2017-2019 **Faculty advisor**, *ACM-W chapter at WWU*, ACM-ICPC Programming Club, Faculty mentor.
- 2014-2016 **Intern supervisor**, *Undergraduate, M.S. and Ph.D. students from University of Hamburg, University of Illinois Urbana-Champaign, Spelman College, University of California San Diego*, NSF REU Advisor, Marshalls College, University of California Berkeley.
- 2011-2013 **Student mentor**, *Undergrad, M.S., Ph.D. students*, Purdue University.